

5. Forensic scientists analyse substances found at crime scenes. They use analytical techniques to arrive at qualitative conclusions. These techniques are being used by students to identify samples of an unknown powder.

The table below shows the effect of adding sodium hydroxide solution to selected cations.

Cation	Effect of adding sodium hydroxide solution
aluminium (Al^{3+})	white precipitate soluble in excess sodium hydroxide to give a colourless solution
ammonium (NH_4^+)	no precipitate when heated with sodium hydroxide, ammonia gas is given off
calcium (Ca^{2+})	white precipitate insoluble in excess sodium hydroxide
copper(II) (Cu^{2+})	blue precipitate insoluble in excess sodium hydroxide
iron(II) (Fe^{2+})	green precipitate insoluble in excess sodium hydroxide
iron(III) (Fe^{3+})	reddish-brown precipitate insoluble in excess sodium hydroxide
lead(II) (Pb^{2+})	white precipitate soluble in excess sodium hydroxide to give a colourless solution
zinc (Zn^{2+})	white precipitate soluble in excess sodium hydroxide to give a colourless solution
magnesium (Mg^{2+})	white precipitate insoluble in excess sodium hydroxide

Tests for some anions are shown in the table below.

Anion	Test	Result
carbonate (CO_3^{2-})	add hydrochloric acid	carbon dioxide gas is produced
chloride (Cl^-)	acidify with dilute nitric acid, then add silver nitrate solution	white precipitate formed
iodide (I^-)	acidify with dilute nitric acid, then add silver nitrate solution	yellow precipitate formed
nitrate (NO_3^-)	add sodium hydroxide solution, then aluminium and warm carefully	ammonia gas produced
sulfate (SO_4^{2-})	acidify with dilute hydrochloric acid and then add barium chloride solution	white precipitate formed
bromide (Br^-)	acidify with dilute nitric acid then add silver nitrate solution	cream precipitate formed

- (a) Describe the difference between qualitative and quantitative analysis. [2]

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- (b) The students dissolve samples of the unknown powder in water and test for anions and cations.

The observations from the students' tests are shown below.

Test	Observations
sodium hydroxide solution added	white precipitate which did not dissolve as more sodium hydroxide was added
acidify then add barium chloride solution	white precipitate
hydrochloric acid added	bubbles of gas
dilute nitric acid then silver nitrate solution added	white precipitate

Use the information in the tables above and opposite to determine:

- (i) all the possible cations that may be present. [1]

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- (ii) all the anions that may be present. [1]

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- (c) Describe another test that could be used to confirm the cation(s) present. [2]

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- (d) Name **one** compound that could be present in the solution and use the symbols given in the tables to write down its chemical formula. [2]

Name

Formula